

# Neehar Peri

neeharperi.com

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## EDUCATION

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|--------------------------------------------------------------------------------------------------------------|----------|
| <b>Ph.D in Robotics, Carnegie Mellon University</b><br><i>Towards Scalable Open-World 3D Perception</i>      | May 2026 |
| <b>M.S in Robotics, Carnegie Mellon University</b><br><i>Long-Tailed 3D Detection via Multi-Modal Fusion</i> | May 2023 |
| <b>B.S. in Computer Engineering, University of Maryland - College Park</b><br><i>QUEST Honors Program</i>    | May 2021 |

## WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>California Institute of Technology</b> , Pasadena, CA, <i>Postdoctoral Research Fellow</i>                                                                                                                                                                                                                                                                                                                                                               | <i>Jun 2026 – Present</i>   |
| <ul style="list-style-type: none"><li>Leading research on vision-language action models and interactive simulators with video diffusion models for contact-rich manipulation</li><li>Advisor: <a href="#">Georgia Gkioxari</a></li></ul>                                                                                                                                                                                                                    |                             |
| <b>RAI Institute</b> , Cambridge, MA, <i>Research Scientist Intern</i>                                                                                                                                                                                                                                                                                                                                                                                      | <i>Jun 2025 – July 2026</i> |
| <ul style="list-style-type: none"><li>Benchmarked SLAM and SfM methods for UMI-style data collection with motion capture data</li><li>Led research on policy learning with multi-modal representations (e.g. depth, force, tactile) for manipulation</li></ul>                                                                                                                                                                                              |                             |
| <b>Carnegie Mellon University</b> , Pittsburgh, PA, <i>Graduate Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                      | <i>Apr 2020 – Apr 2026</i>  |
| <ul style="list-style-type: none"><li>Led research on 3D object detection, multi-object tracking, motion forecasting, and multi-agent planning for embodied perception</li><li>Advisor: <a href="#">Deva Ramanan</a></li></ul>                                                                                                                                                                                                                              |                             |
| <b>NVIDIA</b> , Pittsburgh, PA, <i>Research Scientist Intern</i>                                                                                                                                                                                                                                                                                                                                                                                            | <i>Jan 2024 – Dec 2024</i>  |
| <ul style="list-style-type: none"><li>Led research on 3D object detection in-the-wild</li><li>Built GNN-based tracker that outperforms production system by 5% HOTA and achieves a 10x speedup</li></ul>                                                                                                                                                                                                                                                    |                             |
| <b>MUKH Technologies</b> , College Park, MD, <i>Research Scientist Intern</i>                                                                                                                                                                                                                                                                                                                                                                               | <i>Aug 2020 – May 2023</i>  |
| <ul style="list-style-type: none"><li>Led research on improving thermal-to-visible face synthesis for zero-shot identification</li><li>Built robust face verification pipelines for multi-spectral data streams</li></ul>                                                                                                                                                                                                                                   |                             |
| <b>Argo AI</b> , Pittsburgh, PA, <i>Research Scientist Intern</i>                                                                                                                                                                                                                                                                                                                                                                                           | <i>May 2021 – Oct 2022</i>  |
| <ul style="list-style-type: none"><li>Developed end-to-end 3D object detection and forecasting pipeline from LiDAR point clouds</li><li>Implemented novel metrics that jointly evaluate detection and forecasting accuracy</li></ul>                                                                                                                                                                                                                        |                             |
| <b>University of Maryland</b> , College Park, MD, <i>Undergraduate Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                   | <i>May 2018 – May 2021</i>  |
| <ul style="list-style-type: none"><li>Conducted research in unsupervised traffic anomaly detection and discriminative representation learning for vehicle re-id</li><li>Led research in defending against clean-label adversarial poisoning attacks</li><li>Established novel method for encoding human preferences in revenue maximizing auction design</li><li>Advisors: <a href="#">Rama Chellappa</a> &amp; <a href="#">John P. Dickerson</a></li></ul> |                             |

## AWARDS

| Name                                                         | Institution | Distinction   | Year         |
|--------------------------------------------------------------|-------------|---------------|--------------|
| CVPR Doctoral Consortium                                     | CMU         | International | 2026         |
| NSF Graduate Research Fellowship                             | CMU         | National      | 2023         |
| Maryland Undergraduate Researcher of the Year                | UMD         | University    | 2021         |
| Sujan Guha Memorial Best Senior Thesis Award                 | UMD         | Department    | 2021         |
| CRA Outstanding Undergraduate Researcher (Honorable Mention) | UMD         | National      | 2021         |
| Yurie & Jeong H. Kim Scholarship                             | UMD         | Department    | 20{18,19,20} |

## CONFERENCE PUBLICATIONS

- [DetPO: In-Context Learning with Multi-Modal LLMs for Few-Shot Object Detection](#) ECCV 2026  
*GR Gare\*, N Peri\*, M Popov, SP Jain, J Galeotti, D Ramanan*
- [UniFlow: Zero-Shot LiDAR Scene Flow for Autonomous Driving](#) ECCV 2026  
*S Li, Q Zhang, I Khatri, K Vedder, E Eaton, D Ramanan, N Peri*
- [RefAV: Towards Planning-Centric Scenario Mining](#) CVPR 2026<sup>††</sup> (Award Candidate)  
*C Davidson, D Ramanan, N Peri*
- [RF-DETR: Neural Architecture Search for Real-Time Detection Transformers](#) ICLR 2026  
*I Robinson, P Robicieux, M Popov, D Ramanan, N Peri*
- [Roboflow100-VL: A Multi-Domain Object Detection Benchmark for Vision-Language Models](#) NeurIPS 2025  
*P Robicieux\*, M Popov\*, A Madan, I Robinson, J Nelson, D Ramanan, N Peri*
- [MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion](#) ICCV 2025  
*Z Wang, J Tan, T Khurana\*, N Peri\*, D Ramanan*
- [Towards Learning to Complete Anything in LiDAR](#) ICML 2025  
*A Tacmaz, C Saltori, N Peri, T Meinhardt, RD Lutio, L Leal-Taixe, A Osep*
- [Planning with Adaptive World Models for Autonomous Driving](#) ICRA 2025  
*AB Vasudevan, N Peri, J Schneider, D Ramanan*
- [Neural Eulerian Scene Flow Fields](#) ICLR 2025  
*K Vedder, N Peri, I Khatri, S Li, E Eaton, M Kocamaz, Y Wang, Z Yu, D Ramanan, J Pehserl*
- [Revisiting Few-Shot Object Detection with Vision-Language Models](#) NeurIPS 2024  
*A Madan\*, N Peri\*, S Kong, D Ramanan*
- [Shelf-Supervised Cross-Modal Pre-Training for 3D Object Detection](#) CoRL 2024  
*M Khurana\*, N Peri\*, J Hays, D Ramanan*
- [I Can't Believe It's Not Scene Flow!](#) ECCV 2024  
*I Khatri\*, K Vedder\*, N Peri, D Ramanan, J Hays*
- [Better Call SAL: Towards Learning to Segment Anything in LiDAR](#) ECCV 2024  
*A Osep\*, T Meinhardt\*, F Ferroni, N Peri, D Ramanan, L Leal-Taixe*
- [ZeroFlow: Scalable Scene Flow via Distillation](#) ICLR 2024  
*K Vedder, N Peri, N Chodosh, I Khatri, E Eaton, D Jayaraman, Y Liu, D Ramanan, J Hays*
- [Towards Long-Tailed 3D Detection](#) CoRL 2022  
*N Peri, A Dave, D Ramanan\*, S Kong\**
- [A Brief Survey of Person Recognition at a Distance](#) ASILOMAR 2022  
*C Nalty\*, N Peri\*, J Gleason\*, CD Castillo, S Hu, T Bourlai, R Chellappa*
- [Forecasting from LiDAR via Future Object Detection](#) CVPR 2022  
*N Peri, J Luieten, M Li, A Osep, L Leal-Taixe, D Ramanan*
- [Assessment of a Novel Virtual Environment for Examining Human Cognitive-Motor Performance during Execution of Action Sequences](#) HCII 2022  
*AA Shaver\*, N Peri\*, R Mezebish, G Matthew, A Berson, C Gaskins, GP Davis, GE Katz, I Samuel, JA Reggia, J Purtilo, RJ Gentili*
- [A Synthesis-Based Approach for Thermal-to-Visible Face Verification](#) FG 2021  
*N Peri, J Gleason, CD Castillo, T Bourlai, VM Patel, R Chellappa*
- [PreferenceNet: Encoding Human Preferences in Auction Design with Deep Learning](#) NeurIPS 2021  
*N Peri\*, MJ Curry\*, S Dooley, JP Dickerson*
- [The Devil is in the Details: Self-Supervised Attention for Vehicle Re-ID](#) ECCV 2020  
*P Khorramshahi\*, N Peri\*, JC Chen, R Chellappa*
- [A Dual Path Model with Adaptive Attention for Vehicle Re-ID](#) ICCV 2019<sup>††</sup>  
*P Khorramshahi, A Kumar, N Peri, SS Rambhatla, JC Chen, R Chellappa*

## JOURNAL PUBLICATIONS

- [Accelerating Image Recognition Using High Performance Computing](#) ITEA 2023  
*J Adams, JM Barton, R Chellappa, J Gabberty, J Gleason, S Hu, J Johnson, F Moor-Clingenpeel, B Oshiro, N Peri, D Richie, V To*
- [Data and Algorithms for End-to-End Thermal Spectrum Face Verification](#) TBIOM 2023  
*T Bourlai, J Rose, S Mokalla, A Zabin, L Hornak, CB Nalty, N Peri, J Gleason, CD Castillo, VM Patel, R Chellappa*

## WORKSHOP PUBLICATIONS

- [QuickDraw: Fast Visualization, Analysis and Active Learning for Medical Image Segmentation](#) HCHI 2025  
*D Syomichev\*, P Gopinath\*, GL Wei, E Chang, I Gordon, A Seifu, R Pemmaraju\*, N Peri\*, J Purtilo\**
- [Semi-Supervised Federated Multi-Organ Segmentation with Partial Labels](#) AAPM 2024<sup>††</sup>  
*R Pemmaraju\*, N Peri\**
- [An Empirical Analysis of Range for 3D Object Detection](#) ICCV 2023<sup>††</sup>  
*N Peri, M Li, B Wilson, YX Wang, J Hays, D Ramanan*
- [ReBound: An Open-Source 3D Bounding Box Annotation Tool for Active Learning](#) CHI 2023<sup>†</sup>  
*W Chen\*, A Edgley\*, R Hota\*, J Liu\*, E Schwartz\*, A Yizar\*, N Peri\*, J Purtilo\**
- [Deep k-NN Defense Against Clean-label Data Poisoning Attacks](#) ECCV 2020<sup>†</sup>  
*N Peri\*, N Gupta\*, WR Huang\*, L Fowl, C Zhu, S Feizi, T Goldstein, JP Dickerson*
- [Towards Real-Time Systems for Vehicle Re-ID, Multi-Camera Tracking, and Anomaly Detection](#) CVPR 2020<sup>†</sup>  
*N Peri\*, P Khorramshahi\*, SS Rambhatla\*, V Shenoy, S Rawat, JC Chen, R Chellappa*
- [Attention Driven Vehicle Re-ID and Unsupervised Anomaly Detection for Traffic Understanding](#) CVPR 2019<sup>†</sup>  
*P Khorramshahi, N Peri, A Kumar, A Shah, R Chellappa*

## PREPRINTS

- [Bridging Handheld and Teleoperated Supervision for Contact-Rich Manipulation via State-Gated Experts](#) Under Review  
*V Surendran, N Peri, D Watkins*
- [MimicAgent: Learning Quadruped Skills via Text-to-Trajectory Generation](#) Under Review  
*NP Parameswaran\*, LK Nayak\*, N Peri, D Ramanan*
- [Long-Tailed 3D Detection via Multi-Modal Late Fusion](#) Under Review  
*Y Ma\*, N Peri\*, A Dave, W Hua, D Ramanan, S Kong*

\*Equal Contribution

\*Equal Supervision

<sup>†</sup>Selected for Spotlight Presentation

<sup>††</sup>Selected for Oral Presentation

## INVITED TALKS

- [Foundational Few-Shot Object Detection Challenge](#) Jun 2026  
*Invited Talk: CVPR 2026, Workshop on Open World Vision*
- [Argoverse 2 Scenario Mining Challenge](#) Jun 2025  
*Invited Talk: CVPR 2025, Workshop on Autonomous Driving*
- [Foundational Few-Shot Object Detection Challenge](#) Jun 2025  
*Invited Talk: CVPR 2025, Workshop on Visual Perception via Learning in an Open World*
- [3D Object Detection for Autonomous Vehicles](#) Apr 2025  
*Guest Lecture: 16-825, Learning for 3D Vision*
- [Towards Foundation Models for 3D Perception](#) Mar 2025  
*Invited Talk: Carnegie Mellon University (FLAME Seminar & NeuroAI Seminar)*
- [Image Processing from a Frequency Perspective](#) Feb 2025  
*Guest Lecture: 16-720, Computer Vision*
- [Long-Tailed 3D Detection via 2D Late Fusion](#) Oct 2024  
*Invited Talk: ECCV 2024, Workshop on Vision-Centric Autonomous Driving*
- [Shelf-Supervised Cross-Modal Pre-Training for 3D Object Detection](#) Oct 2024  
*Invited Talk: ECCV 2024, Autonomous Vehicles meet Multimodal Foundation Models Workshop*
- [Argoverse 2 End-to-End Forecasting Challenge](#) Jun 2024  
*Invited Talk: CVPR 2024, Workshop on Autonomous Driving*
- [Foundational Few-Shot Object Detection Challenge](#) Jun 2024  
*Invited Talk: CVPR 2024, Workshop on Visual Perception via Learning in an Open World*
- [3D Object Detection for Autonomous Vehicles](#) Apr 2024  
*Guest Lecture: 16-720, Computer Vision*
- [Better Call SAL: Towards Learning to Segment Anything in LiDAR](#) Apr 2024  
*Invited Talk: Stack AV*
- [3D Object Detection for Autonomous Vehicles](#) Apr 2024  
*Guest Lecture: 16-825, Learning for 3D Vision*

- [Long-Tailed 3D Object Detection via Multi-Modal Fusion](#) Jan 2024  
*Invited Talk: Carnegie Mellon University (R-PAD Lab)*
- [An Empirical Analysis of Range for 3D Object Detection](#) Oct 2023  
*Invited Talk: ICCV 2023, Robustness and Reliability of Autonomous Vehicles in the Open-World*
- [Argoverse 2 End-to-End Forecasting Challenge](#) Jun 2023  
*Invited Talk: CVPR 2023, Workshop on Autonomous Driving*
- [3D Object Detection for Autonomous Vehicles](#) Mar 2023  
*Guest Lecture: 16-825, Learning for 3D Vision*
- [Image Processing and Convolutions](#) Sep 2022  
*Guest Lecture: 16-720, Computer Vision*
- [How do Autonomous Vehicles See the World?](#) Aug 2022  
*Invited Talk: Carnegie Mellon University (RoboLaunch)*
- [Transformers for Vision](#) Apr 2022  
*Guest Lecture: 16-720, Computer Vision*
- [Training Convolutional Neural Networks](#) Apr 2022  
*Guest Lecture: 16-720, Computer Vision*
- [Metrics and Methods for Detection and Forecasting in Autonomous Vehicles](#) Apr 2022  
*Invited Talk: National Autonomous Vehicle Conference*

## MENTORSHIP

| Name                                                                                                                       | Institution             | Year(s)     | Project                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Narayanan Parameswaran                                                                                                     | CMU                     | 2025 –      | Text-to-policy for quadrupedal locomotion                                                                                                               |
| Lucky Nayak                                                                                                                | CMU                     | 2025 –      | Text-to-policy for quadrupedal locomotion                                                                                                               |
| Shruti Jain                                                                                                                | CMU                     | 2025        | Benchmarking spatial understanding in VLMs                                                                                                              |
| Siyi Li                                                                                                                    | UPenn                   | 2025 –      | Zero-shot scene flow estimation via cross-domain generalization, VLM post-training for spatial understanding                                            |
| Cainan Davidson                                                                                                            | CMU                     | 2024 –      | Benchmarking scenario mining for AVs, VLAs for offroad driving                                                                                          |
| Guang-Lin Wei, Eric Chang, Padmini Gopinath, Ian Gordon, Amanuel Seifu, Daniel Syomichev                                   | UMD                     | 2024        | CMSC435 software engineering capstone to build an <a href="#">active-learning framework</a> for medical image analysis                                  |
| Zihan Wang                                                                                                                 | CMU → Berkeley PhD      | 2024 – 2025 | Sparse-view dynamic reconstruction in-the-wild                                                                                                          |
| Nina Johe, Aryan Kakadia, Muzzamil Khan, Morgan Ko, Josh Leeman, Max Son, Sashwat Venkatesh                                | UMD                     | 2024        | CMSC435 software engineering capstone to build an end-to-end platform for medical image analysis                                                        |
| Mehar Khurana                                                                                                              | IIITD → CMU MSR         | 2023 – 2024 | Shelf-supervised 3D object detection with VLMs                                                                                                          |
| Anish Madan                                                                                                                | CMU → SkildAI           | 2022 – 2025 | Few-shot multi-modal 2D detection with VLMs                                                                                                             |
| Andrew Shen                                                                                                                | CMU → OpenAI            | 2022 – 2023 | Benchmarking modular 3D perception stack for AVs                                                                                                        |
| Aminah Yizar, Andrew Edgley, Ezra Schwartz, Joshua Liu, Raunak Hota, Royce He, Wesley Chen                                 | UMD                     | 2022        | CMSC435 software engineering capstone to build an <a href="#">active learning framework</a> to allow human-in-the-loop 3D object annotation             |
| Christopher Nalty                                                                                                          | MUKH → Oregon State PhD | 2021 – 2022 | Synthetic data augmentation for thermal-to-visible face verification                                                                                    |
| Aastha Senjalia, Andrew Vetter, Benjamin Namovicz, Cheyenne Montgomery, Ferzam Mohammad, Matthew Weinberg, Nicholas Revill | UMD                     | 2021        | CMSC435 software engineering capstone to build a <a href="#">visualization platform</a> for autonomous vehicle data. Project won People’s Choice Award. |

## PROFESSIONAL SERVICE

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**Conference Reviewer:** NeurIPS, CVPR, AAAI, ICCV, ICLR, ECCV, ICRA, CoRL

**Journal Reviewer:** IJCV, TPAMI

**Mentorship:** CMU AI Mentoring Program, QUEST Mentoring Program, CMU AI for Social Good Summit

**Masters Thesis Committee Member:** Bharath Raj, Anish Madan

**Other:** AUTOBOT & TRINITY Cluster Management, RISS Admissions Committee, CMU Computer Vision Reading Group

## ORGANIZED WORKSHOPS & CHALLENGES

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| Event                                                                   | Venue | Year |
|-------------------------------------------------------------------------|-------|------|
| 6th Workshop on Open-World Vision (OWV)                                 | CVPR  | 2026 |
| RF20-VL Few-Shot Object Detection Challenge @ OWV Workshop              | CVPR  | 2026 |
| Scenario Mining Challenge @ Workshop on Autonomous Driving              | CVPR  | 2026 |
| Scene Flow Challenge @ Workshop on Autonomous Driving                   | CVPR  | 2026 |
| 5th Workshop on Visual Perception and Learning in an Open World (VPLOW) | CVPR  | 2025 |
| RF20-VL Few-Shot Object Detection Challenge @ VPLOW Workshop            | CVPR  | 2025 |
| Scenario Mining Challenge @ Workshop on Autonomous Driving              | CVPR  | 2025 |
| Scene Flow Challenge @ Workshop on Autonomous Driving                   | CVPR  | 2025 |
| 4th Workshop on Visual Perception and Learning in an Open World (VPLOW) | CVPR  | 2024 |
| nuImages Few-Shot Object Detection Challenge @ VPLOW Workshop           | CVPR  | 2024 |
| End-to-End Forecasting Challenge @ Workshop on Autonomous Driving       | CVPR  | 2024 |
| 3rd Workshop on Visual Perception and Learning in an Open World (VPLOW) | CVPR  | 2023 |
| End-to-End Forecasting Challenge @ Workshop on Autonomous Driving       | CVPR  | 2023 |
| 2nd Workshop on Visual Perception and Learning in an Open World (VPLOW) | CVPR  | 2022 |

## TEACHING EXPERIENCE

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**16-720**, Carnegie Mellon University, Robotics Institute, *Head Teaching Assistant*

*Spring 2022, Fall 2022*

- Managed team of teaching assistants to effectively coordinate course responsibilities
- Graded course projects and held office hours

**ENEE 244**, University of Maryland, ECE Department, *Undergraduate Teaching Fellow*

*Spring 2019*

- Led Introduction to Digital Logic recitation for a discussion section of 15 students